

Effective Communication for Selection Processes in Computing: An Experience Report

Charles N C Freitas¹ , Gabriel A M Fragoso¹ , Pedro H A A Barros¹ , André L P Silva¹  and Gabriel R S Silva¹ 

¹ Centro de Informática, UFPE, Recife, Brazil
{cncf, gamf, phaab, alps2, grss}@cin.ufpe.br

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Abstract: Effective communication has become an essential competency for computing professionals, particularly in recruitment and selection processes, where interpersonal skills complement technical expertise. However, undergraduate computing programs often emphasize technical knowledge while providing limited opportunities for communication skills development. This paper presents an experience report on a four-hour minicourse designed to prepare computing students for job interviews. The intervention was structured using the ADDIE instructional design model and followed a Human-AI-Technique-Human pedagogical sequence, integrating problematization with a Human Resources specialist, AI-mediated mock interviews, instruction on the STAR method and storytelling, and peer role-playing activities. A mixed-methods approach was adopted, combining diagnostic surveys, performance data collected through the Winspeak.ai platform, and post-intervention evaluations. Based on experience with 18 participants, the results showed high participant satisfaction (score of 4.76 out of 5), increased self-confidence, progressive improvement during AI-mediated interview simulations, and recognition of the STAR method as a valuable framework for structuring interview responses. The main challenges identified were eloquence, response organization, and communication under pressure. The findings suggest that integrating artificial intelligence, structured communication strategies, and collaborative learning provides a promising approach to developing communication skills and better preparing computing students for recruitment processes and the labor market.

¹ <https://orcid.org/0009-0002-2996-7138>

² <https://orcid.org/0009-0004-0448-2884>

³ <https://orcid.org/0000-0003-3606-8827>

⁴ <https://orcid.org/0009-0002-0529-9155>

⁵ <https://orcid.org/0000-0000-0000-0000>

1 INTRODUCTION

Communication has become a core competency for computing professionals, particularly during recruitment processes, where candidates must demonstrate not only technical expertise but also the ability to explain, justify, and contextualize their decisions to recruiters, teams, and non-technical stakeholders (Amarasena et al., 2024; Meier and Laumer, 2022).

Despite this relevance, undergraduate computing curricula rarely offer structured opportunities to develop these competencies, prioritizing technical content over interpersonal and communicative training (Ivory et al., 2024; Oluwagbeminiyi, 2025). This gap is particularly visible among early-career candidates, who, despite adequate technical preparation, often lack experience in professional interviews and self-presentation, which affects their ability to demonstrate their actual potential during selection processes (Naing et al., 2025; Shivaji, 2025; Diniz et al., 2025).

Recent advances in Artificial Intelligence, particularly Large Language Models (LLMs) and Natural Language Processing, offer a promising path to address this gap: AI-mediated tools can simulate interview scenarios and provide personalized, low-risk feedback at scale, complementing traditional human-led training (Kansada, 2025; MR et al., 2025; Dhivya et al., 2025).

Motivated by this opportunity, we designed and evaluated a four-hour minicourse, *Effective Communication for Selection Processes*, structured around a progressive pedagogical architecture that combines human problematization, AI-mediated practice, structured communication instruction, and peer role-playing. This paper reports the design, implementation, and evaluation of this intervention, with the aim of contributing to the discussion on complementary, AI-supported strategies for closing the communication-skills gap in computing education.

The remainder of this paper is organized as follows. Section 2 reviews related work on communication skills, employability, and AI-mediated training. Section 3 describes the institutional context and the diagnostic that motivated the intervention. Section 4 details the course design and pedagogical rationale. Section 5 presents the methodology used to evaluate the intervention. Section 6 reports the results. Section 7 discusses their implications, and Section 8 concludes the paper.

2 BACKGROUND AND RELATED WORK

2.1 Communication Skills and Employability in Computing

Communication skills are increasingly associated with employability and career progression in technology fields, since technical knowledge alone is insufficient when professionals must explain reasoning and translate contributions into organizational value (Amarasena et al., 2024; Meier and Laumer, 2022; Diniz et al., 2025). However, computing education has traditionally prioritized technical content over the systematic development of these competencies, producing what the literature describes as a soft-skills gap (Ivory et al., 2024; Johnson et al., 2025).

This gap becomes especially visible during recruitment, where candidates must explain their reasoning and communicate confidence under pressure (Naing et al., 2025; Rao et al., 2025). The literature reports that the resulting difficulties are multidimensional: evaluative anxiety reduces fluency and impairs logical organization (Mongkoljaturong et al., 2025; Nirgude and Awasekar, 2025); inexperienced candidates struggle to self-assess their own performance, limiting the value of unsupervised practice (Ohba et al., 2022; Pereira and Hone, 2021); and candidates often struggle to externalize technical reasoning verbally or to organize past experiences into coherent narratives (Daryanto et al., 2025; Lyons, 2025). Communication training for computing students must therefore address not only what candidates say, but how they organize, deliver, and adapt their answers.

2.2 AI-Mediated Tools and Structured Communication Strategies

AI-mediated tools can simulate interview scenarios and provide individualized feedback at scale, reducing the emotional cost of repeated practice (Naing et al., 2025; Dhivya et al., 2025; MR et al., 2025). However, the literature cautions that such tools are not full replacements for human evaluators: automated systems may struggle to assess complex technical reasoning or to respond empathically to deep-seated anxiety, and concerns remain regarding privacy and algorithmic bias (Johnson et al., 2025; Nirgude and Awasekar, 2025). AI is therefore best understood as a complementary mechanism for

scalable, low-risk practice rather than a substitute for human feedback.

Structured frameworks help candidates organize what they communicate. The STAR method (Situation, Task, Action, Result) guides candidates to structure behavioral answers around a logical sequence of context, responsibility, intervention, and impact, while storytelling techniques help connect experiences and outcomes into a more engaging narrative (Lyons, 2025). The literature converges on the recommendation that effective preparation should combine conceptual instruction, guided practice, iterative feedback, and social interaction, with AI-mediated simulation supporting individualized practice and human-led activities supporting interpretation, empathy, and peer learning (Johnson et al., 2025; Luo and Li, 2025; Nirgude and Awasekar, 2025).

The minicourse described in this report responds directly to this recommendation. Rather than treating AI, structured frameworks, and human interaction as independent resources, we organize them into a single, ordered pedagogical sequence: Human-AI-Technique-Human, in which each component prepares participants for the next. Section 3 introduces the diagnostic that motivated this design, and Section 4 details its instructional rationale.

3 COURSE CONTEXT

3.1 Institutional Context and Design Process

The intervention was developed as the final project of a graduate course on Educational Innovations in Computing, in which student teams design, implement, and evaluate educational solutions. The team, five graduate students acting jointly as instructional designers and facilitators, adopted the ADDIE model (Analysis, Design, Development, Implementation, Evaluation) as the methodological reference for the design process, given its iterative structure for aligning educational objectives, pedagogical strategies, and learning outcomes (Peterson, 2003).

The resulting product, a four-hour, in-person minicourse titled "Effective Communication for Selection Processes in Computing," was offered as an extracurricular activity, since the undergraduate computing programs involved do not include formal,

credit-bearing components dedicated to communication training or interview preparation. The course did not seek to replace existing instructional practices, but to address a formative gap identified during the Analysis stage of ADDIE. Of 30 available seats, 27 students registered and 18 effectively participated (a 67% attendance rate).

3.2 Learner Diagnosis

To characterize this formative gap empirically and to establish a baseline for measuring learning gains, a diagnostic survey was administered to all 27 registered students prior to the minicourse. Respondents were predominantly men (70.4%) between 18 and 30 years old (89%); 22 were Computer Science students, and the remainder were enrolled in related technology programs. Regarding prior interview experience, 55.6% had participated in one to three selection processes, 37.0% had never participated in any, and only 7.4% had been approved for an internship — a profile consistent with an early-career population with limited practical interview experience.

As shown in Table 1, nervousness and anxiety were the most frequently reported obstacle (17 of 27 respondents), followed by difficulty highlighting personal differentials and insecurity in technical interviews (11 respondents each), and difficulty structuring answers and communicating prior experience (9 respondents each).

Table 1 — Most frequently reported communication difficulties (diagnostic survey, n = 27).

Reported difficulty (n = 27)	Frequency
Nervousness / anxiety	17
Difficulty highlighting personal differentials	11
Difficulty in technical interviews	11
Difficulty structuring answers	9
Difficulty talking about own experience	9
Never participated in an interview	7

A complementary self-assessment, in which participants rated their competency across seven behaviors on a five-point Likert scale, reinforced this picture (Table 2). The highest-rated competency was receiving feedback and adjusting communication accordingly (mean 3.63), suggesting a positive predisposition toward learning. The

lowest scores were observed for communicating under pressure (2.70) and identifying one's own professional strengths (2.78) — competencies directly tied to high-stakes, evaluative situations such as interviews.

Table 2 — Self-assessed competency by behavior, 1–5 scale (diagnostic survey, $n = 27$).

Self-assessed competency (1–5 scale)	Mean
Receiving feedback and adjusting communication	3.63
Speaking clearly about one's own trajectory	3.15
Confidence for interviews	3.15
Structuring behavioral answers	2.89
Presenting experience objectively	2.78
Identifying own professional strengths	2.78
Communicating under pressure	2.70

Together, the diagnostic survey and self-assessment indicate that participants' main challenges were not technical but communicative, socio-emotional, and reflective, concentrated precisely on performance under evaluative pressure. This finding directly shaped the pedagogical architecture presented in Section 4: since anxiety under evaluation was the dominant barrier, the course needed to introduce evaluative exposure gradually rather than immediately confronting participants with high-pressure human interaction.

4 COURSE DESIGN

4.1 Pedagogical Architecture

To address the gradual-exposure requirement identified in Section 3.2, the minicourse content was organized around two complementary frameworks, the STAR method, which structures behavioral answers around situation, task, action, and result (Lyons, 2025), and storytelling, which contributes narrative coherence and engagement, combined with a deliberate sequencing of human and AI-mediated interaction.

This sequencing is the central instructional design contribution of this work and is summarized as:

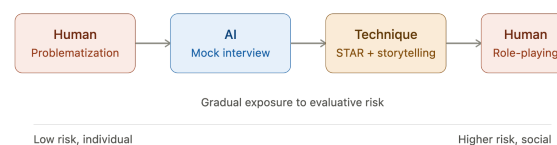


Figure 1 — Conceptual architecture of the minicourse.

Participants begin with a human interaction conducted by an HR specialist, who contextualizes the central problem, effective communication in job interviews. Critically, participants then engage in an AI-mediated interview simulation *before* receiving any formal instruction on communication frameworks. This ordering is intentional: it captures a more authentic baseline of participants' spontaneous communicative strategies, uninfluenced by recently learned content or rehearsed answers. Only after this baseline is captured does structured instruction on STAR and storytelling take place, followed by a peer role-playing activity in which participants apply what they have learned in a human interaction context.

This sequence operationalizes a strategy of gradual exposure to evaluative risk: each stage increases the level of social exposure and perceived evaluation, moving participants from an individual, low-risk, AI-mediated environment toward higher-exposure human interaction, directly addressing the evaluative anxiety identified as the dominant barrier in Section 3.2.

4.2 Instructional Flow

The intervention was structured as a single four-hour, in-person session organized into seven sequential blocks (Figure 2), implementing the Human-AI-Technique-Human sequence described above.

The session opens with goal framing (10 min), followed by the Problematization block (50 min), in which an HR specialist shares real recruitment experiences through group discussion and Q&A, contextualizing the problem and establishing an authentic scenario for subsequent activities.

Participants then individually complete an AI-mediated interview simulation (60 min) using Winspeak.ai, which scores responses from 0 to 100 across five dimensions: confidence, social intelligence, clarity, storytelling, and eloquence. As discussed in Section 4.1, this activity precedes any formal technique instruction by design.

The Technique Instruction block (45 min) then introduces the STAR method and storytelling, using worked examples that contrast strong and weak interview answers, with an emphasis on

immediate applicability rather than theoretical exposition. After a 15-minute break, participants engage in peer Role-Playing (30 min): organized into trios, they rotate through three 15-minute cycles (10-minute simulated interview plus 5-minute peer feedback), so that every participant experiences both the candidate's and the interviewer's perspective. The session closes with a guided Experience Evaluation (30 min), combining self-assessment with a closing discussion.



Figure 2 — Instructional flow of the minicourse (4-hour session, single cohort).

4.3 Pedagogical Principles

The instructional strategy integrates four complementary pedagogical principles. Problem-Based Learning anchors the Problematization block in an authentic recruitment scenario, grounding subsequent activities in a real professional challenge. Microlearning shapes the Technique Instruction block, concentrating STAR and storytelling into a focused 45-minute session built around immediately applicable, comparative examples rather than extended theoretical treatment. Collaborative learning is operationalized through the systematic role rotation in the Role-Playing block, which exposes every participant to both the candidate's and the interviewer's perspective, deepening their understanding of evaluation criteria. Finally, gradual exposure to evaluative risk, the pedagogical element that most distinguishes this intervention from conventional lecture-then-practice

models, governs the overall sequencing described in Section 4.1, postponing direct human judgment until participants have first practiced individually in a low-risk, AI-mediated setting.

Facilitation followed a two-tier model: two instructors led content delivery and session coordination, while two tutors supported participants individually during practical activities and administered the assessment instruments. The invited HR specialist participated only in the Problematization block, contributing professional authenticity to the initial discussion.

5 METHODOLOGY

This section outlines the methodological framework adopted for data collection and analysis, consolidating the academic rigor required to evaluate the instructional intervention's empirical impact. To mitigate inherent self-assessment biases and achieve an objective triangulation of findings, the research design relied on a mixed-methods approach, articulating automated telemetry logs with qualitative perceptions gathered directly from the participants.

5.1 Research Design

The intervention was evaluated as an intrinsic single-case study, using a pre-test/post-test design without a control group, to investigate the effectiveness of the Human-AI-Technique-Human sequence in developing transversal communicative competencies under compressed temporal conditions. Data capture and analysis followed the Evaluation phase of the ADDIE model, keeping curriculum design, execution, and outcome measurement methodologically aligned.

5.2 Instruments

The target population comprised undergraduate computing students. Data capture was strategically organized around three complementary instruments designed to encompass the behavioral, cognitive, and affective dimensions of the learning experience:

1. **Baseline diagnostic survey** (described in Section 3.2), administered at enrollment, mapping demographic profile, prior interview experience, and self-reported confidence and anxiety.
2. **Winspeak.ai telemetry**, recorded automatically during the individual AI simulation: session counts, performance scores (0–100) across five dimensions (clarity, confidence, eloquence, social intelligence, storytelling), and qualitative feedback strings generated for each attempt.
3. **Post-intervention experience evaluation**, administered after the role-playing activity, measuring facilitator quality, simulator usability, and Net Promoter Score (NPS), and re-applying the pre-test confidence item to allow direct pre/post comparison.

5.3 Data Analysis

The data analysis protocol followed distinct pipelines for quantitative and qualitative data streams to synthesize the empirical evidence. Within the quantitative pipeline, the baseline diagnostic metrics, the performance scores from the graded AI attempts, and the experience evaluation scales were processed using descriptive and inferential statistics. The direct intervention impact on the primary objective was measured by calculating the absolute perceptual gain, contrasting the initial baseline confidence against the final post-intervention score. Furthermore, the longitudinal learning curve driven by repeated practice was modeled by computing a linear trend and calculating the mean slope of score progression across successive practices for all users with multiple completed simulation sessions.

Consequently, the qualitative textual data underwent a rigorous process of thematic categorization and coding. The extensive volume of automated feedback text logs extracted from Winspeak.ai was categorized according to the platform's core dimensions, quantifying the frequency of positive mentions against targeted improvement points to identify systemic behavioral patterns across the cohort. In parallel, the

open-ended text comments from the post-intervention survey were examined through inductive thematic analysis. This approach allowed the research team to map out the primary learning outcomes consolidated by the students, identify systematic development gaps, and isolate actionable opportunities for the long-term refinement of subsequent instructional design iterations.

6 RESULTS

This section presents the results of the intervention based on two main sources: usage and performance records from the individual AI-mediated simulation conducted through Winspeak.ai and the experience evaluation questionnaire completed at the end of the minicourse. The first source refers exclusively to the individual AI-mediated stage, which took place before the formal instruction on the STAR method and storytelling. The second source encompasses the entire minicourse experience, including conceptual instruction, use of the simulator, and the final peer role-playing activity.

6.1 AI Simulation: Usage and Performance

During the Winspeak.ai-mediated stage, 17 students completed at least one individual mock interview practice. In total, 62 practice sessions were initiated, of which 57 received a complete evaluation. Among the students who used the platform, the average was 3.65 practices per participant, suggesting that once students engaged with the tool, a substantial portion performed multiple attempts.

The average score across evaluated attempts was 47.0 (median 40.0, scale 0–100). Among the five dimensions assessed by the platform, the highest average scores were observed in confidence (56.5), social intelligence (50.6), and clarity (50.5), whereas storytelling (41.3) and eloquence (36.0) obtained the lowest scores, Figure 3 shows the average AI simulation score by communication dimension.

This pattern suggests that participants demonstrated some degree of confidence and interpersonal ability but still faced difficulties in

structuring more elaborate responses, building narratives, and adapting their communication to the professional standards typically expected in job interviews.

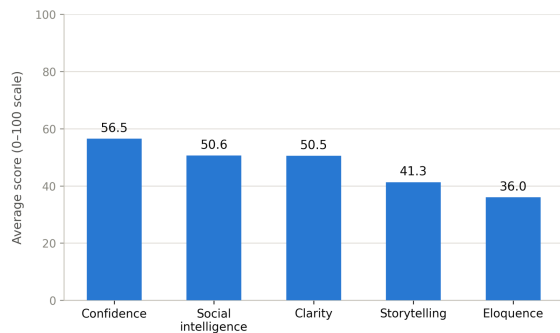


Figure 3 — Average AI simulation score by communication dimension (0–100 scale).

6.2 Learning Outcomes

The post-intervention questionnaire received 17 responses (94.4% of the 18 active participants). Self-reported confidence in structuring and communicating responses rose from 2.94 to 3.76 (1–5 scale), as shown in Figure 4; 12 of 17 respondents reported increased confidence, three remained stable, and two reported a decrease.

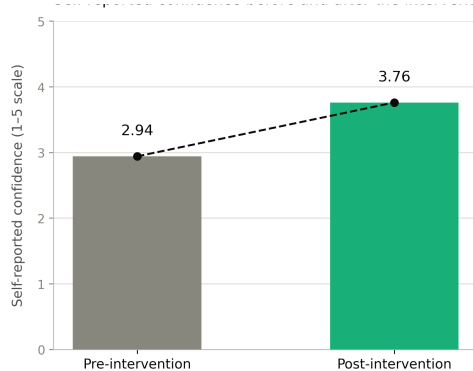


Figure 4 — Self-reported confidence before and after the intervention (1–5 scale).

Among open-ended responses, the STAR method was the most frequently cited learning outcome, particularly for organizing responses and reporting experience. Assertive communication, conciseness, and interview preparation were also

recurrent themes. However, 10 of 17 respondents reported only partial understanding of the concepts, indicating a need for continued practice beyond the single session.

The peer role-playing activity was positively received: 16 of 17 respondents reported that rotating roles helped them better understand interview evaluation criteria, and 13 considered the allotted time adequate, though open-ended feedback converged on a desire for more simulation and feedback time in future editions.

6.3 Course Evaluation

The Figure 5 shows that overall satisfaction averaged 4.76 (1–5 scale), with facilitator clarity (4.65), material quality (4.88), and group activity usefulness (4.65) all rated highly. The recommendation score (NPS basis) averaged 9.47 (0–10, median 10); 14 of 17 respondents scored 9 or 10, and no participant scored below 7.

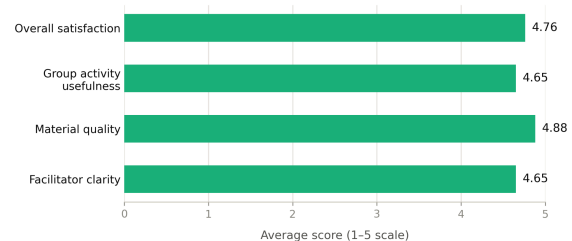


Figure 5 — Course evaluation summary.

Simulator usability was similarly favorable: ease of use (4.59), quick learnability (4.53), willingness to reuse (4.47), and confidence while using the platform (4.18) all scored highly, while negative usability indicators (complexity, inconsistency, difficulty) scored below 2. These results suggest that Winspeak.ai was perceived as an accessible tool well suited to the minicourse context. Nevertheless, the item related to the need for assistance when using the simulator obtained an intermediate average score, indicating that some participants could benefit from more explicit initial guidance. In future applications, a brief guided demonstration before individual practice may reduce

entry barriers and foster greater autonomy in the use of the platform.

6.4 Summary of Results

Taken together, the results indicate that the minicourse was well received by participants and produced positive evidence both in terms of observed performance and perceived learning. The individual AI-mediated simulation generated a sufficient volume of practice sessions to identify patterns of performance, progression, and formative feedback, while the final questionnaire revealed high satisfaction, strong recommendation rates, and increased self-reported confidence.

The findings also reinforce the coherence between the initial diagnosis and the intervention outcomes. The difficulties previously identified, namely anxiety, response structuring, and communication of experiences, reappeared both in the platform feedback and in students' perceptions. At the same time, the appreciation of the STAR method and the peer role-playing activity suggests that participants recognized the practical value of the proposed strategies, particularly when applied to simulated situations that closely resemble real-world selection processes.

7 DISCUSSION

The intervention demonstrated that a minicourse focused on effective communication addresses a critical formative gap in computing education. While students often possess high technical proficiency, they face significant socio-emotional barriers during selection processes, particularly when communicating under pressure.

7.1 Effectiveness of the Pedagogical Architecture

The defining feature of this experience was the Human-AI-Technique-Human model. By positioning the AI simulation before formal instruction, the team captured an authentic baseline of student skills without the influence of rehearsed

answers. This approach allowed for a gradual exposure to evaluative risk. Since nervousness and anxiety were the primary difficulties identified (reported by 17 out of 27 students), the low-risk and private environment of the AI simulator helped mitigate these emotional barriers before the higher-pressure peer interactions.

7.2 The Role of AI and the STAR Method

Data from the Winspeak.ai platform revealed that while students had reasonable clarity, eloquence (36.0) and storytelling (41.3) were their weakest areas. The average improvement of 3.96 points per attempt suggests that repetitive practice with automated feedback facilitates practical learning. Furthermore, the STAR method was the most frequently highlighted learning outcome. This indicates that technical students respond well to logical and systematic structures for organizing professional narratives that might otherwise feel subjective or difficult to frame.

7.3 Perceptions and Limitations

The increase in self-reported confidence from 2.94 to 3.76 and the high recommendation rate of 9.47/10 underscore the relevance of the intervention. However, the development of soft skills is an ongoing process. The fact that 10 out of 17 respondents reported only a partial understanding of the concepts indicates that a single four-hour session is effective for diagnosis and initial awareness but is insufficient for full competency consolidation. Participant feedback explicitly called for more time for simulation and feedback, which suggests a need for longitudinal approaches.

7.4 Implications for Computing Education

The results confirm that computing education must extend beyond technical code. The difficulty students faced in communicating personal strengths and performing under pressure highlights the need for pedagogical spaces that use Problem-Based Learning (PBL) and collaborative learning to simulate market demands. The high usability scores

for the AI tool suggest that such technologies are scalable allies for instructors who seek to integrate transversal competencies into technical curricula.

8 CONCLUSIONS

This experience report detailed the implementation of a communication minicourse for computing students using an innovative Human-AI-Technique-Human pedagogical sequence. The intervention proved to be an effective response to the formative gap identified in computing programs, which frequently lack formal curricular components dedicated to communicative competencies.

The findings indicate that the use of AI-mediated simulations provided a safe and low-risk environment for students to confront their primary hurdle, which was identified as interview-related anxiety. The objective improvement in performance scores alongside the subjective gain in student confidence, which increased from a mean of 2.94 to 3.76, suggest that the combination of immediate feedback and structured techniques like the STAR method is highly effective for short-term skill acquisition.

However, the study also identified that a single four-hour session serves only as a starting point for professional development. While the participants highly valued the experience, they expressed a clear desire for additional practice time and deeper simulation cycles. Therefore, future work should focus on integrating these activities more permanently into the computing curriculum or expanding the minicourse into a more extensive longitudinal program. Further research is also recommended to determine if these gains in confidence and technique translate effectively into higher success rates during actual job market placement over the long term.

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